

Application Protocols: TELNET, FTP

FALL 2021

CSC311

PROF. EARL



TELNET [RFC 854, 855]

RFC published in 1983. The original version predates the TCP/IP Internet (ARPANet)

Features:

- Allows for logging into any type of terminal from any type of computer
- Establishes TCP connection from one site to another
- TELNET Server generally listens on port 23 (IANA Standard)
- Client/Server protocol.
- 8-Bit ASCII characters are supported.

Established connections use a NVT (Network Virtual Terminal)

- A Keyboard and a Printer
- Allows for options to be set between client and server.

Client accepts keyboard characters from users and relays them **one at a time** to the Server.

- Server 'echoes' received characters back to client.

FTP – File Transfer Protocol [RFC 959]

RFC published in 1985

Goal was to create an easy way to share programs/files between computers reliably and efficiently

Designed to be used primary by programs, but also can be used in a terminal.

Uses client/server model. Use of **two parallel** TCP connections.

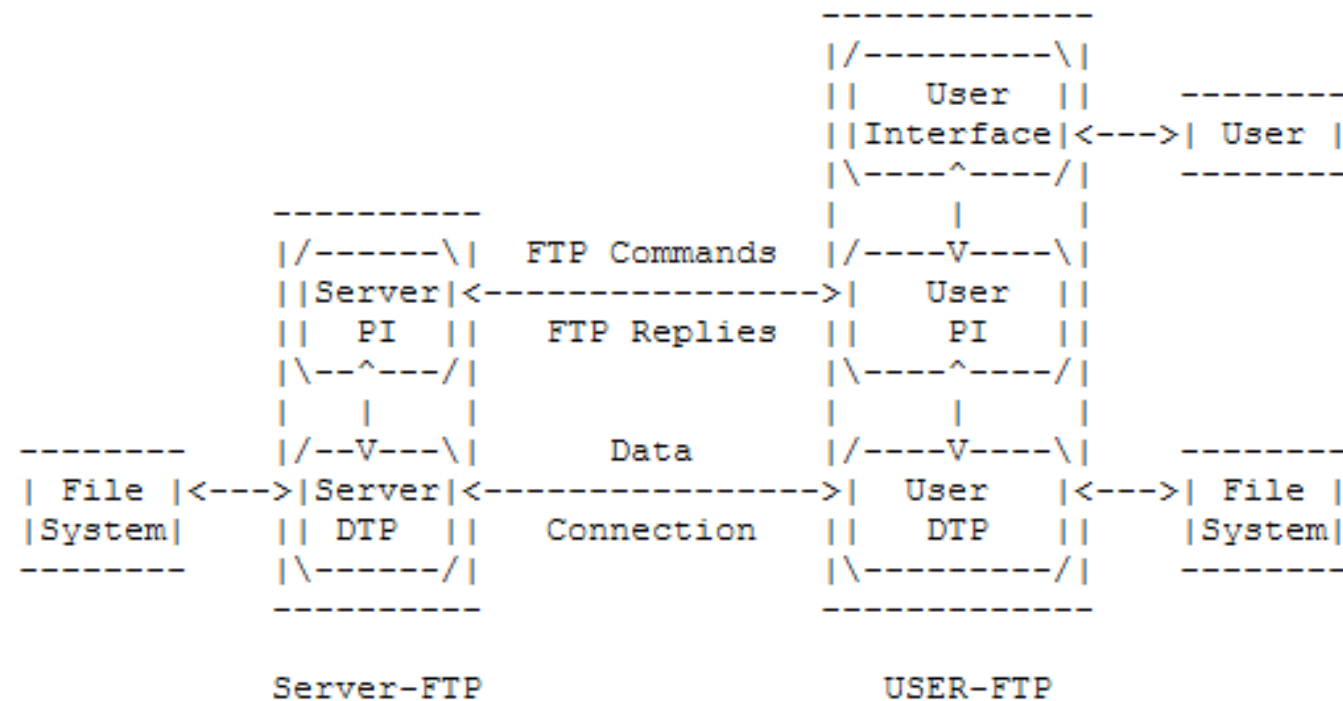
Control Connection

- Transfer of commands and replies
- Follows telnet protocol
- Out-of-Band

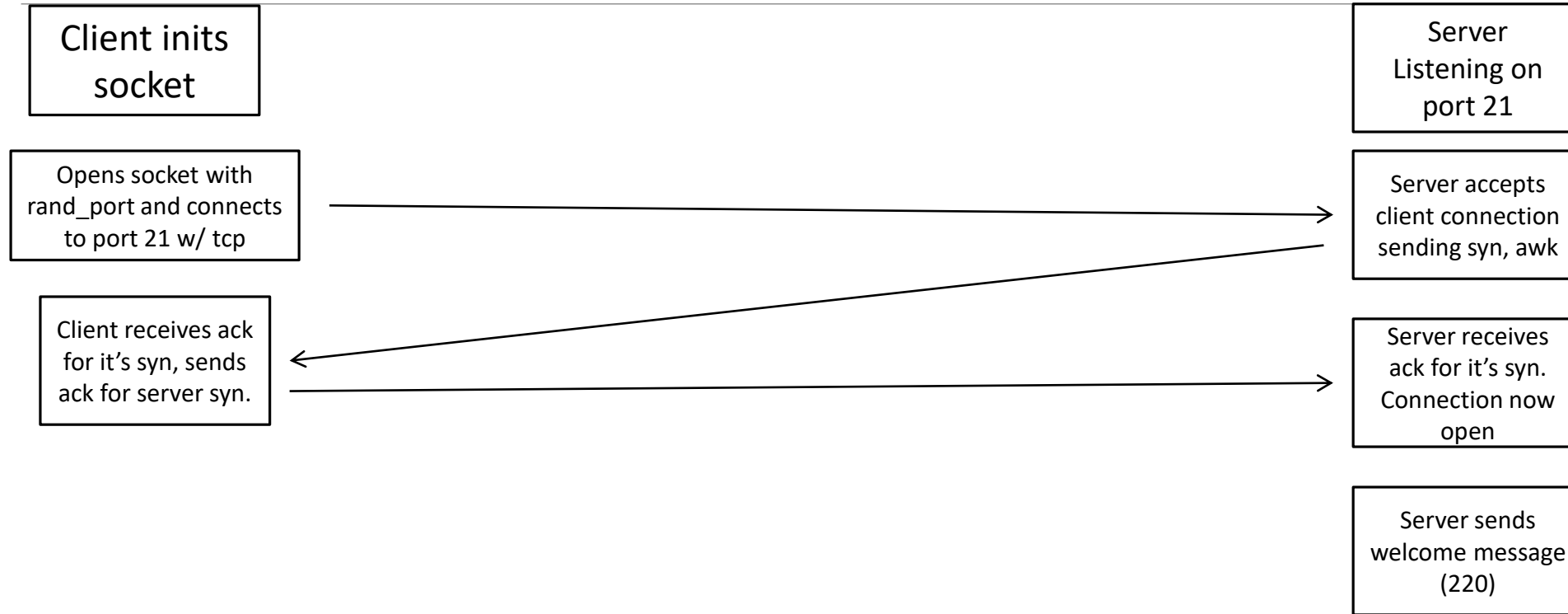
Data Connection

- Used for transfer of data
- Full-Duplex
- In-Band

The FTP Model

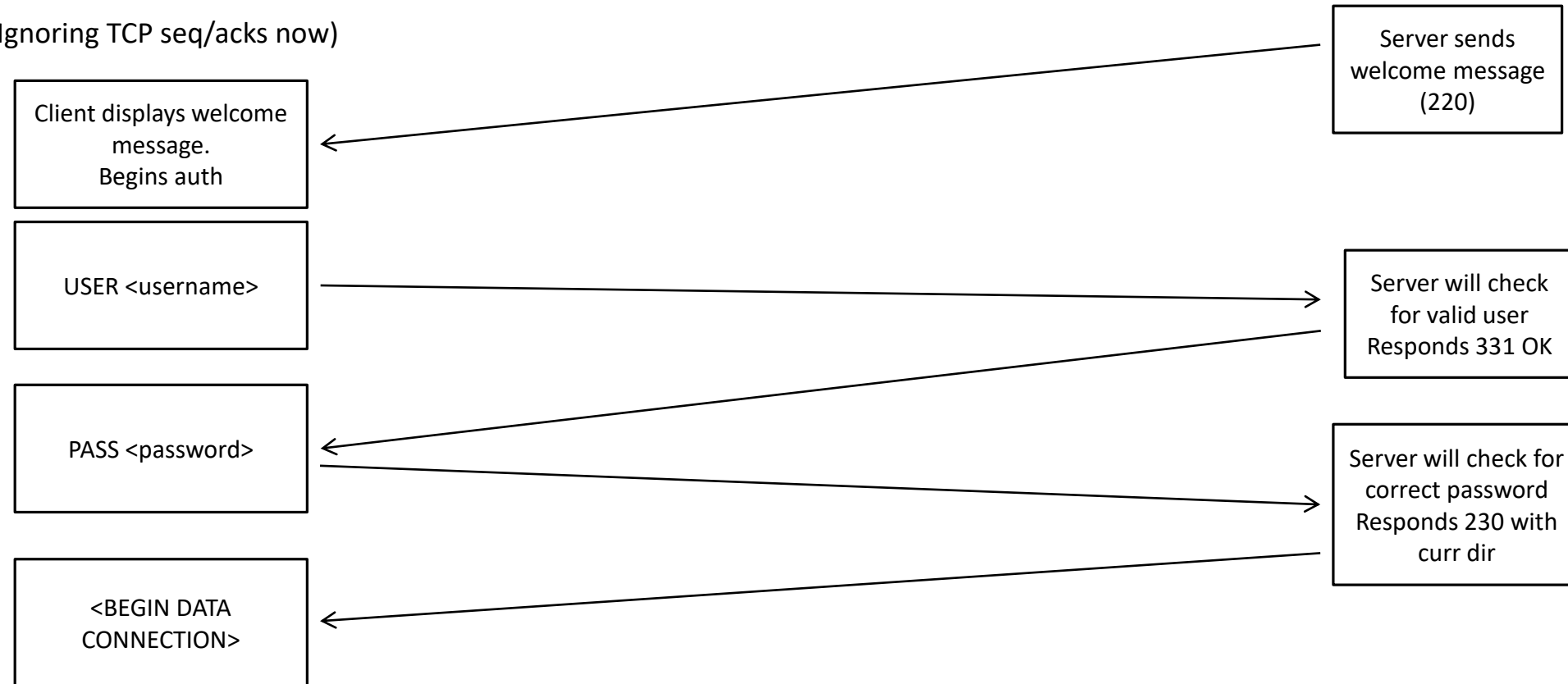


FTP Command Connection Setup



FTP Command Connection Setup

(Ignoring TCP seq/acks now)



FTP Data Modes – Passive and Active

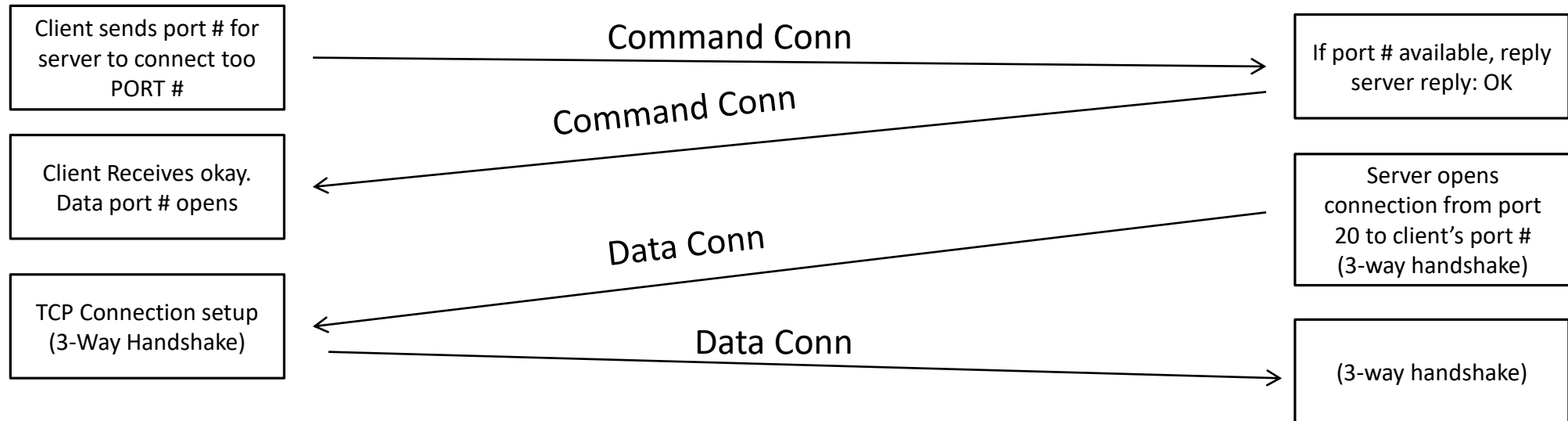
Active Mode – Older Mode.

- Client initiates connection from random port # (x) to FTP server on Port 21
- Client sends PORT command with a port # (y)
- Server connects to specified port # on client through its own port 20.
- TCP Conn X → 21 for *Control Connection*
- TCP Conn 20 → Y for *Data Connection*

Passive Mode

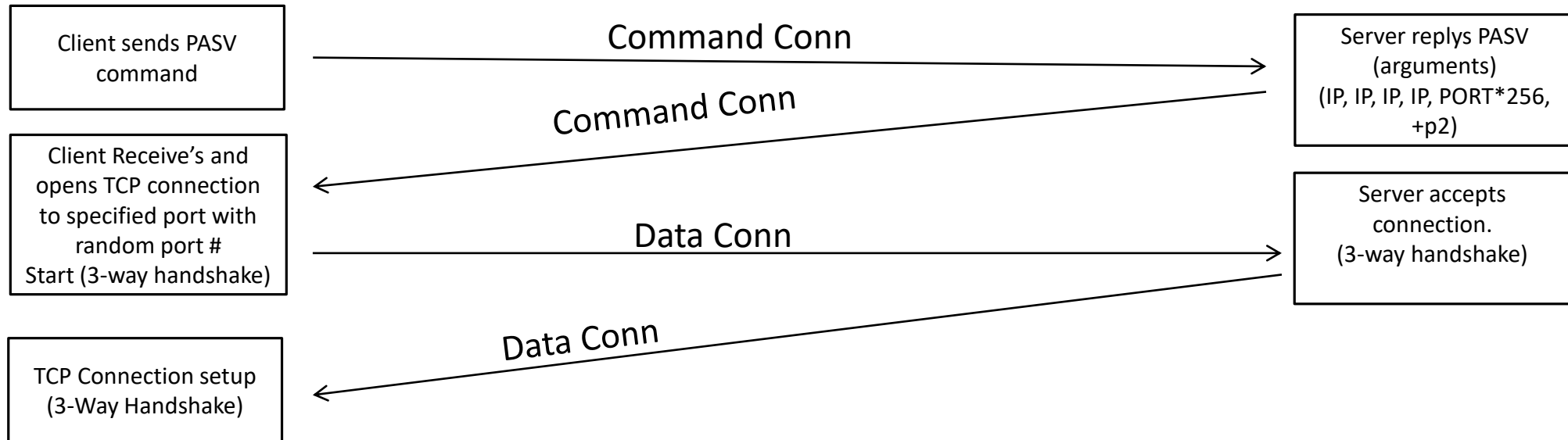
- Client still initiates connection from random port # (x) to FTP server
- Client instead sends PASV command
- Server responds with random port # (y) for client to connect to.
- TCP Conn X → 21 for *Control Connection*
- TCP Conn rand_port → Y *Data Connection*

FTP Active Mode



File transfer can now start

FTP Passive Mode



File transfer can now start